

HAIJIAN SUN

Contact Information

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Research Interests

AI-Native Wireless and Learning Systems: Data-Driven and Physics-Informed Modeling of Wireless Channel; Generative Waveform and Neural Transceiver Design; Learning-Enabled PHY/MAC Optimization for Adaptive Communication and Sensing.

Wireless Intelligence and Edge Autonomy: Multimodal Data Fusion for Perception and Communication; Edge AI for Real-Time Decision Making; Trustworthy and Energy-Efficient Communication in Large-Scale IoT and Cyber-Physical Systems.

Experimental Systems and Prototyping: Software-Defined Radio Platforms, O-RAN Testbeds, and AI-Ready Wireless Datasets for Validation and Open Research.

Education

Utah State University, Logan, UT, USA *Sept. 2014 - Jul. 2019*
Ph.D. in Electrical and Computer Engineering
Thesis: *Spectral, Energy and Computation Efficiency in Future 5G Wireless Networks*
Ph.D. Scholar of the Year, 2019.
Advisor: [Rose Qingyang Hu](#) (Now VT ECE Head)

Xidian University, Xi'an, China *Sept. 2011 - May 2014*
M.S. in Electrical Engineering (Telecommunication)

Xidian University, Xi'an, China *Sept. 2007 - Aug. 2011*
B.S. in Electrical Engineering (Telecommunication)

Professional Experience

University of Georgia Athens, GA
Assistant Professor of ECE *Aug. 2022 - Present*

University of Wisconsin-Whitewater Whitewater, WI
Assistant Professor of CS *Aug. 2019 - Jul. 2022*

Mitsubishi Electric Research Laboratories (MERL) Cambridge, MA
Research Intern *May 2018 - Aug. 2018*

Honors and Awards

- [A1] **UGA Emerging Scholar Research Award**, 2026.
Recognize UGA College of Engineering's Scholarly Excellence among Assistant Professors
- [A2] **IEEE ISICN 2024, Best Paper Award**, 2024.

- [A3] **NSF Platforms for Advanced Wireless Research “Find a Rover Challenge” Award**, 2024.
- [A4] **NSF CISE Research Initiation Initiative (CRII) Award**, 2022.
Early-Career initiative award in NSF Directorate for Computer and Information Science and Engineering (CISE)
- [A5] **NSF Engineering Research Initiative (ERI) Award**, 2022.
Early-Career initiative award in NSF Directorate for Engineering (ENG)
- [A6] **Chancellor’s Innovation Award for Computer Science: UW-Whitewater**, 2022.
Presented in 2022 State of the University Address
- [A7] **Regent Scholar Award: University of Wisconsin (UW) System**, 2021.
The most prestigious award for faculty in undergraduate research excellence across UW campuses
- [A8] **Ph.D. of the Year: Utah State University**, 2019.
Outstanding Engineering Ph.D. Scholar of 2019
- [A9] **ESI Highly Cited Papers: Clarivate**, 2019, 2018.
Top 1% in Wireless Communication Journal Publications
- [A10] **National Scholarship: Ministry of Education of the People’s Republic of China**, 2012.
The highest honor for graduate students in China
- [A11] **Distinguished Scholarship: Xidian University** in 2012, awarded to top 5% students

Funded Research

Extramural

Total: \$22.25 million; UGA (since 2022): \$7.4 million; My share: \$4.28 million

- [F1] **Collaborative Research: VINES: Track 2: SCOPE-XR: Seeing the Channel: Optimizing Perception-aware Environments for XR**

Lead-PI, National Science Foundation, total: \$5,100,000, UGA (Lead) share: \$2,167,500, with UT-Austin, UW-Madison, Virginia Tech, and Ericsson Inc. 08/2026 - 07/2029.
- [F2] **Collaborative Research: CCSS: Rethinking Wireless Channel Through the Lens of Radiance Field**

Lead-PI, National Science Foundation, total: \$480,000, UGA (Lead) share: \$280,000, with Virginia Tech. 09/2025 - 08/2028.
- [F3] **Collaborative Research: CISE Core Small: NeTS: Intelligent Reflecting Surface Assisted Physical Layer Security Enhancement for Ultra-dense IoT Networks**

PI, National Science Foundation, total: \$600,000, UGA share: \$225,000. 10/2025 - 09/2028.
- [F4] **POSE: Phase I: SpectraGuru: Building an Open-Source Ecosystem for Raman and Spectroscopic Research and Education**

Lead-PI, National Science Foundation, \$300,000. 08/2025 - 07/2026.
- [F5] **Tribal & Rural Autonomous Vehicles for Equity, Livability and Safety (TRAVELS)**

Co-PI, US Department of Transportation, total: \$15,000,000, UGA \$3,600,000, my share: ~\$720,000 (Lead: UW-Madison). 2025 - 2031.
- [F6] **CNS: CRII: Towards Spectrum and Energy Efficient Large-scale IoT Communications: A Cross-layer Optimization Approach**

Sole-PI, National Science Foundation, \$229,998 (with supplemental). 05/2023 - 04/2026.

[F7] **NSF Advance Wireless Platform AERPAW “Find A Rover Challenge” Award**

Sole-PI, National Science Foundation, \$2,000. Unconstrained Gift Award. 01/2024.

[F8] **ECES: ERI: Toward mmWave Vehicular Communication: A Multisensor Multimodal Deep Data Fusion Approach**

Sole-PI, National Science Foundation, \$199,955. 06/2022 - 05/2024.

[F9] **mmWave Dual Channel Device (28 GHz)**

Sole-PI, Tmytek Inc. Device Gift Donation, \$20,000. Date: 11/2022.

[F10] **A Multimodal Deep Sensor Fusion System for Reliable and Faster Next-Gen Wireless Vehicle Communications**

Lead-PI, University of Wisconsin System Regent Scholar Award, total: \$50,000, my share: \$40,000. 01/2021 - 08/2022.

[F11] **Privacy Protection among Community-Dwelling Older Wisconsinites in the Age of Internet of Things**

Co-PI, Thompson Center, University of Wisconsin System, total: \$105,757, my share: \$35,252. 07/2021 - 06/2022.

[F12] **Developing Prototype Multimodal Sensors and Wireless Network System for Landslide Monitoring and Early Warning**

Co-PI, University of Wisconsin System Regent Scholar Award, total: \$50,000, my share: \$16,666. 01/2020 - 08/2021.

[F13] **Dual-Antenna RFID Reader Prototype Development for RFID Chip-Based Vehicle Localization**

Lead-PI, Sponsor: Tokef LLC (private company), total: \$112,051.00, my share: \$56,026, 07/2020 - 08/2021.

Intramural

[I1] **Data-driven Ultra-fast, Reliable, and Low-latency Wireless Communication for E-Vehicles**

PI, UGA E-Mobility Initiative, \$56,000. Project Date: 7/2023 - 6/2024.

[I2] **Data-Driven Distributed Communication, Control, and Computing**

PI, UGA Pre-seed Grant for Interdisciplinary Research, \$4,000. Project Date: 12/2022 - 12/2023.

[I3] **A Precision Tracking and Tracing System in Food Supply Chain Through Ubiquitous Wireless Connectivity**

Sole-PI, IIPA UGA, \$37,500. Project Date: 11/2022 - 6/2023.

[I4] **Enabling Smart Farming Through Interconnectivity Infrastructure Systems**

PI (25%), IIPA UGA, \$245,000. Project Date: 11/2022 - 11/2023.

[I5] **Research Catalyst Grant**

Sole-PI (100%), Office of Research and Sponsored Program, University of Wisconsin-Whitewater, \$10,000. 01/2021 - 12/2021.

Google Scholar: <https://scholar.google.com/citations?user=nHVq0B8AAAAAJ&hl=en>

Books/Chapters

- [B1]. **H. Sun**, R. Q. Hu, and Y. Qian, “5G and Beyond Wireless Communication Networks”, ISBN:9781119089506, Wiley, 2024. [Book Link in Publisher Website](#)
- [B2]. Z. Zhang, **H. Sun**, X. Lei, “Non-orthogonal Multiple Access”, Book Chapter, in *Encyclopedia of Wireless Networks*, 2018.

White Papers/Roadmap

- [W1]. **H. Sun**, C. T. K. Ng, J. Huo, R. Q. Hu, N. Wang, and C. M. Chen, “IEEE Beyond 5G Technology Roadmap – Massive MIMO Group”, 1st Edition, *IEEE Future Networks*, 2019. [Link](#)

Journal Publications

- [J1]. * H. Zhang and **H. Sun**, “Feasibility Guaranteed Learning-to-Optimize in Wireless Communication Resource Allocation”, *IEEE Wireless Communications*, 2026.
- [J2]. **H. Sun** and F. Ye, “Radio Radiance Field: The New Frontier of Spatial Wireless Channel Representation”, to appear, *IEEE Communications Magazine*, 2026.
- [J3]. * L. Zhang, **H. Sun**, S. Berweger, C. Gentile, and R. Q. Hu, “RF-3DGS: Wireless Channel Modeling with Radio Radiance Field and 3D Gaussian Splatting”, *IEEE Transactions on Wireless Communication*, 2026.
- [J4]. S. Masrur *et al.*, “Collection: UAV-Based RSS Measurements from the AFAR Challenge in Digital Twin and Real-World Environments”, *IEEE Data Descriptions*, 2025.
- [J5]. X. Le, Y. Sun, **H. Sun**, R. Q. Hu, D. Niyato, and MA. Imran, “Joint Power and Spectrum Orchestration for D2D Semantic Communication Underlying Energy-Efficient Cellular Networks”, *IEEE Transactions on Wireless Communication*, 2025.
- [J6]. H. Guo, Y. Ye, **H. Sun**, L. Shi, and R. Q. Hu, “Resource Allocation for Mutualistic Symbiotic Radio with Hybrid Active-Passive Communications”, *IEEE Transactions on Cognitive Communications and Networking*, 2025.
- [J7]. * H. Zhang, M. Chen, A. Vahid, F. Ye, and **H. Sun**, “Model-based Deep Learning for QoS-Aware Rate-Splitting Multiple Access Wireless Systems”, *IEEE Transactions on Wireless Communication*, 2025.
- [J8]. R. Xu, Y. Ye, **H. Sun**, L. Shi, and G. Lu, “Revolutionizing Symbiotic Radio: Exploiting Tradeoffs in Hybrid Active-Passive Communications”, *IEEE Communications Magazine*, 2025.
- [J9]. Y. Yang, X. Xu, Y. Zeng, **H. Sun**, and R. Q. Hu, “Channel Knowledge Map for Cellular-Connected UAV via Binary Bayesian Filtering”, *IEEE Transactions on Communications*, 2025.
- [J10]. H. Yu, Y. Liu, Z. Yang, **H. Sun**, and M. Chen, “Optimizing Wireless Resource Management and Synchronization in Digital Twin Networks”, *IEEE Internet of Things Journal*, 2025.
- [J11]. S. Lu, Y. Ye, **H. Sun**, L. Shi, and R. Q. Hu, “Minimizing Total Transmission Time in Hybrid Active-Passive Mutualistic Symbiotic Radio”, *IEEE Wireless Communications Letters*, 2025.
- [J12]. * X. Ma, **H. Sun**, R. Q. Hu, and Y. Qian, “Approximate Wireless Communication for Lossy Gradient Updates in IoT Federated Learning”, *IEEE Internet of Things Journal*, 2024.
- [J13]. A. Vahid, **H. Sun**, S-C. Lin, “Movable Antenna-Aided Broadcast Packet Erasure Channels: Capacity with Dynamic Position Plan”, *IEEE Wireless Communication Letters*, 2024.

- [J14]. C. Liu, J. Wang, L. Feng, H. Lu, **H. Sun**, and R. Q. Hu, “Coalitional Game Theory-Based Resource Allocation Strategy for Robust IRS-VLC System”, *Photonics*, 2024.
- [J15]. W. Zhou, L. Shi, X. Song, **H. Sun**, and G. Zheng, “Performance Analysis for Relay Assisted Short-Packet Backscatter Communications”, *IEEE Internet of Things Journal*, 2024.
- [J16]. X. Song, D. Han, L. Shi, **H. Sun**, and R. Q. Hu, “Relay Assisted Cooperative Ambient Backscatter Communication with Hybrid Long-Short Packets”, *IEEE Transactions on Vehicular Technology*, 2024.
- [J17]. * X. Ma, Y. Zhou, H. Zhang, Q. Wang, **H. Sun**, H. Wang, R. Q. Hu, “Exploring Communication Technologies, Standards, and Challenges in Electrified Vehicle Charging”, *IET Communications*, 2024.
- [J18]. Y. Cang, M. Chen, Y. Pan, Z. Yang, H. Ye, **H. Sun**, M. Chen, “Joint User Scheduling and Computing Resource Allocation Optimization in Asynchronous Mobile Edge Computing Networks”, *IEEE Transactions on Communications*, 2024.
- [J19]. * L. Zhang, **H. Sun**, Y. Zeng, R. Q. Hu, “Spatial Channel State Information Prediction with Generative AI: Towards Holographic Communication and Digital Radio Twin”, *IEEE Network*, 2024.
- [J20]. * E. Zhu, **H. Sun**, and M. Ji, “Physics-informed Generalizable Wireless Channel Modeling with Segmentation and Deep Learning: Fundamentals, Methodologies, and Challenges”, *IEEE Wireless Communications*, 2024.
- [J21]. **H. Sun**, X. Ma, and R. Q. Hu, and R. Christensen, “Precise Coil Alignment for Dynamic Wireless Charging of Electric Vehicles with RFID Sensing”, *IEEE Wireless Communications*, 2024.
- [J22]. T. Nguyen, **H. Sun**, H. Wang, and R. Q. Hu, “Authentication and PHY-Security Schemes for Electric Vehicle Dynamic Wireless Charging”, *IEEE Transactions on Vehicular Technology*, 2023.
- [J23]. R. Xu, L. Shi, Y. Ye, **H. Sun**, G. Zheng, “Relay-Enabled Backscatter Communications: Linear Mapping and Resource Allocation”, *IEEE Transactions on Vehicular Technology*, 2023.
- [J24]. L. Shi, X. Chu, H. Sun and G. Lu, “Wireless Powered OFDMA-MEC Networks with Hybrid Active-Passive Communications,” *IEEE Internet of Things Journal*, 2023.
- [J25]. Z. Liu, Y. Ye, X. Chu and **H. Sun**, “Secrecy Performance of Backscatter Communications with Multiple Self-Powered Tags,” *IEEE Communications Letters*, 2022.
- [J26]. Q. Wang, **H. Sun**, R. Q. Hu, and A. Bhuyan, “When Machine Learning Meets Spectrum Sharing Security: Methodologies and Challenges”, *IEEE Open Journal of the Communications Society*, 2021.
- [J27]. Y. Xu, **H. Sun**, Y. Ye, “Distributed Resource Allocation for SWIPT-based Cognitive Ad-Hoc Networks”, *IEEE Transactions on Cognitive Communications and Networking*, 2021.
- [J28]. **H. Sun**, X. Ma, and R. Q. Hu, “Adaptive Federated Learning With Gradient Compression in Uplink NOMA”, *IEEE Transactions on Vehicular Technology*, 2020.
- [J29]. T. Koike-Akino, P. Wang, M. Pajovic, **H. Sun**, and P. V. Orlik, “Fingerprinting-Based Indoor Localization With Commercial MMWave WiFi: A Deep Learning Approach”, *IEEE Access*, 2020.
- [J30]. Y. Ye, L. Shi, **H. Sun**, R. Q. Hu, and G. Lu, “System-Centric Computation Energy Efficiency for Distributed NOMA-Based MEC Networks”, *IEEE Transactions on Vehicular Technology*, 2020.
- [J31]. B. Liu, J. Song, J. Wang, **H. Sun**, Q. Wang, “Robust Secure Wireless Powered MISO Cognitive Mobile Edge Computing”, *IEEE Access*, 2020.

- [J32]. **H. Sun**, F. Zhou, and R. Q. Hu, “Joint Offloading and Computation Energy Efficiency Maximization in a Mobile Edge Computing System”, *IEEE Transactions on Vehicular Technology*, 2019. **ESI Highly Cited Paper!** (top 1% in 2019)
- [J33]. **H. Sun**, F. Zhou, R. Q. Hu, and L. Hanzo, “Robust Beamforming Design in a NOMA Cognitive Radio Network Relying on SWIPT”, *IEEE Journal on Selected Areas in Communications*, 2019. **ESI Highly Cited Paper!** (top 1% in 2019)
- [J34]. **H. Sun**, Z. Zhang, R. Q. Hu, and Y. Qian, “Wearable Communications in 5G: Challenges and Enabling Technologies”, *IEEE Vehicular Technology Magazine*, 2018.
- [J35]. F. Zhou, Z. Chu, **H. Sun**, R. Q. Hu, and L. Hanzo, “Artificial Noise Aided Secure Cognitive Beamforming for Cooperative MISO-NOMA Using SWIPT”, *IEEE Journal on Selected Areas in Communications*, 2018. **ESI Highly Cited Paper!** (top 1% in 2018)
- [J36]. Z. Zhang, **H. Sun** and R. Q. Hu, “Downlink and Uplink Non-Orthogonal Multiple Access in a Dense Wireless Network”, *IEEE Journal on Selected Areas in Communications*, 2017.

Peer-Reviewed Conference Publications

- [C1]. * L. Zhang, P. Kudyba, Z. An, and **H. Sun**, “RadioSight: Predictive mmWave XR Network Optimization from Dynamic Neural Radio Fields”, conditional accept, **ACM MobiCom**, 2026.
- [C2]. C. Xu, H. Zhang, **H. Sun**, and F. Ye, “LLM-supported 3D Modeling Tool for Radio Radiance Field Reconstruction, submitted, *IEEE ICC*, 2025.
- [C3]. * J. Song, L. Zhang, F. Ye, and **H. Sun**, “Terahertz Spatial Wireless Channel Modeling with Radio Radiance Field”, accepted, *IEEE Globecom 2025*.
- [C4]. S. Sabud, **H. Sun**, S. Banerjee, F. Ye, “MuJam-RIS: A DRL Approach for Selective Multi-User Jamming Powered by RIS and Spatial-CSI”, *ACM WiseML 2025*.
- [C5]. S. Hossain, W. Tang, C. Chenli, **H. Sun**, W. Song, S. Lee, M. Bowman, T. Jung, “MtDB: A Decentralized Multi-Tenant Database for Secure Data Sharing”, *IEEE Blockchain 2025*.
- [C6]. * P. Kudyba, **H. Sun**, “Autonomous Agricultural Monitoring with Aerial Drones and RF Energy-Harvesting Sensor Tags”, *IEEE ISICN 2025*.
- [C7]. D. Chen, Q. Wang, **H. Sun**, and Y. Hao, “GNN Based PLS Enhancement for Next Generation Spectrum Sharing Industrial Networks”, *IEEE ISICN 2025*.
- [C8]. * H. Zhang, M. Chen, A. Vahid, F. Ye, and **H. Sun**, “Model-based Deep Learning for Wireless Resource Allocation in RSMA Communications Systems”, *IEEE ICC 2025*.
- [C9]. W. Guo, Q. Wang, H. Yue, **H. Sun**, and R. Q. Hu, “Efficient Phishing URL Detection Using Graph-based Machine Learning and Loopy Belief Propagation”, *IEEE ICC 2025*.
- [C10]. * P. Kudyba, Q. Lu, **H. Sun**, “Bayesian Optimization for Fast Radio Mapping and Localization with an Autonomous Aerial Drone”, *IEEE VTC 2024*.
- [C11]. N. Tahir, R. Parasuraman, **H. Sun**, “Communication-Aware Consistent Edge Selection for Mobile Users and Autonomous Vehicles”, *IEEE VTC 2024*.
- [C12]. * P. Kudyba, **H. Sun**, “Wireless FEC-less Frame Transmission for Model Aggregation of Distributed Learning Systems”, *IEEE INFOCOM (Demo) 2024*.
- [C13]. * E. Zhu, **H. Sun**, M. Ji, “HawkRover: An Autonomous mmWave Vehicular Communication Testbed with Multi-sensor Fusion and Deep Learning”, *IEEE ISICN 2024*. **Best Paper Award!**

- [C14]. * L. Zhang, **H. Sun**, J. Sun, and R. Q. Hu “WiSegRT: Dataset for Site-Specific Indoor Radio Propagation Modeling with 3D Segmentation and Differentiable Ray-Tracing”, *Invited Paper, IEEE ICNC 2024*.
- [C15]. * H. Zhang, **H. Sun**, T. He, W. Xiang, R. Q. Hu, “Energy Efficient Robust Beamforming for Vehicular ISAC with Imperfect Channel Estimation”, *IEEE ICC 2024*.
- [C16]. * L. Zhang, **H. Sun**, J. Sun, R. Parasuraman, Y. Ye, R. Q. Hu, “Map2Schedule: An End-to-End Link Scheduling Method for Urban V2V Communications”, *IEEE ICC 2024*.
- [C17]. H. Guo, Y. Ye, **H. Sun**, L. Shi, “Resource Allocation for Mutualistic Symbiotic Radio with Hybrid Active-Passive Communications”, *IEEE Globecom 2023*.
- [C18]. * X. Ma, **H. Sun**, R. Q. Hu and Y. Qian, “Approximate Wireless Communication for Federated Learning”, *ACM WiseML 2023*.
- [C19]. Y. Zhou, **H. Sun**, X. Ma and R. Q. Hu, “Energy-Efficient Secure Offloading for NOMA-Enabled Machine-type Mobile-Edge Computing,” *IEEE International Conference on Industrial Technology (ICIT) 2023*.
- [C20]. X. Gao, Y. Ye, G. Lu and **H. Sun**, “Throughput Fairness-Aware Optimization of Cognitive Backscatter Networks with Finite Alphabet Inputs,” *IEEE/CIC International Conference on Communications in China (ICCC) 2022*.
- [C21]. X. Ma, **H. Sun**, R. Q. Hu and Y. Qian, “A New Implementation of Federated Learning for Privacy and Security Enhancement,” *IEEE Globecom 2022*.
- [C22]. R. Xu, Y. Ye, **H. Sun** and G. Lu, “Wireless Powered Opportunistic Cooperative Backscatter Communications: To Relay or Not?,” *IEEE VTC 2022*.
- [C23]. X. Ma, **H. Sun**, Q. Wang, and R. Q. Hu, “User Scheduling for Federated Learning Through Over-the-Air Computation”, *IEEE VTC 2021*.
- [C24]. X. Ma, **H. Sun**, and R. Q. Hu, “Scheduling Policy and Power Allocation for Federated Learning in NOMA Based MEC”, *IEEE Globecom 2021*.
- [C25]. B. Steck, O. Yavuzcetin, J. Bhattacharyya, and **H. Sun**, “Develop a Prototype Multimodal Device for Monitoring Slope Instability using Raspberry PI”, *the Geological Society of America Connect 2021* (as a peer-reviewed poster)
- [C26]. Q. Wang, H. Hu, **H. Sun**, and R. Q. Hu, “Secure and Energy-Efficient Offloading and Resource Allocation in a NOMA-Based MEC Network”, *IEEE/ACM Symposium on Edge Computing (SEC) 2020*.
- [C27]. J. Yu, P. Wang, T. Koike-Akino, Y. Wang, P. V. Orlik, and **H. Sun**, “Human Pose and Seat Occupancy Classification with Commercial mmWave WiFi”, *IEEE Globecom 2021*.
- [C28]. J. Yang, Y. Xu, Q. Liu, Q. Li, and **H. Sun**, “Energy-efficient Resource Allocation for Secure IRS Networks with Active Eavesdropper”, *IEEE ICC 2020*.
- [C29]. **H. Sun**, P. Wang, M. Pajovic, T. Koike-Akino, P. V Orlik, A. Taira, and K. Nakagawa, “Fingerprinting-Based Outdoor Localization with 28-GHz Channel Measurement: A Field Study”, *IEEE SPAWC 2020*.
- [C30]. **H. Sun**, Q. Wang, X. Ma, Y. Xu, and R. Q. Hu, “Towards Green Mobile Edge Computing Offloading Systems with Security Enhancement”, *IEEE i-ETC 2020*.
- [C31]. P. Wang, M. Pajovic, T. Koike-Akino, **H. Sun**, and P. Orlik, “Fingerprinting-Based Indoor Localization with Commercial MMWave WiFi – Part I: RSS and Beam Indices”, *IEEE Globecom 2019*.

- [C32]. P. Wang, M. Pajovic, T. Koike-Akino, **H. Sun**, and P. Orlik, “Fingerprinting-Based Indoor Localization with Commercial MMWave WiFi – Part II: Spatial Beam SNRs”, *IEEE Globecom 2019*.
- [C33]. F. Zhou, **H. Sun**, Z. Chu, and R. Q. Hu, “Computation Efficiency Maximization for Wireless-Powered Mobile Edge Computing”, *IEEE Globecom 2019*.
- [C34]. **H. Sun**, F. Zhou, and Z. Zhang, “Robust Beamforming Design in a NOMA Cognitive Radio Network Relying on SWIPT”, *IEEE ICC 2018*.
- [C35]. F. Zhou, Z. Chu, **H. Sun**, and V. C. M. Leung, “Resource Allocation for Secure MISO-NOMA Cognitive Radios Relying on SWIPT”, *IEEE ICC 2018*.
- [C36]. F. Zhou, Y. Wu, **H. Sun**, and Z. Chu, “UAV-Enabled Mobile Edge Computing: Offloading Optimization and Trajectory Design”, *IEEE ICC 2018*.
- [C37]. **H. Sun**, Q. Wang, R. Q. Hu, “Outage Probability in a NOMA Relay System,” *IEEE WCNC*, San Francisco, May 2017.
- [C38]. **H. Sun**, Q. Wang, S. Ahmed and R. Q. Hu, “Non-Orthogonal Multiple Access in a mmWave Based IoT Wireless System with SWIPT,” **invited paper** *IEEE VTC 2017*.
- [C39]. Y. Xu, **H. Sun**, R. Q. Hu, “Hybrid MU-MIMO and Non-orthogonal Multiple Access Design in Wireless Heterogeneous Networks,” *EUSIPCO 2016*.
- [C40]. Z. Zhang, **H. Sun**, R. Q. Hu, “Stochastic Geometry Based Performance Study on 5G Non-Orthogonal Multiple Access Scheme”, *IEEE Globecom 2016*.
- [C41]. **H. Sun**, B. Xie, R. Q. Hu and G. Wu, “Non-orthogonal Multiple Access with SIC Error Propagation in Downlink Wireless MIMO Networks,” **invited paper**, *IEEE VTC 2016*.
- [C42]. **H. Sun**, Y. Xu and R. Q. Hu, “A NOMA and MU-MIMO Supported Cellular Network with Underlaid D2D Communications”, *IEEE VTC 2016*.
- [C43]. Y. Xu, **H. Sun**, R. Q. Hu, and Y. Qian, “Cooperative Nonorthogonal Multiple Access in Heterogeneous Networks,” *IEEE Globecom 2015*.

Non Peer-Reviewed Papers/Posters

- [Po1]. G. Lu, et. al, “AGI for Agriculture”, *arXiv preprint arXiv:2304.06136*.
- [Po2]. F. Dou, et. al, “Towards Artificial General Intelligence (AGI) in the Internet of Things (IoT): Opportunities and Challenges”, *arXiv preprint arXiv:2309.07438*.
- [Po3]. S. Flynn, J. Schnor, **H. Sun**, H. Nguyen, and J. Zhou, “Hawk Rover: An Autonomous Multi-sensor Platform for Next-Gen Wireless Vehicular Communication”, *UWW Undergraduate Research Day Poster*, Oct. 2021.
- [Po4]. A. Jeninga, R. McCarthy, N. Huffman, E. Jeninga, G. Lewis, B. Steck, O. Yavuzcetin, J. Bhattacharyya, and **H. Sun**, “Developing a Prototype Multimodal Device for Landslide Monitoring”, *UWW Undergraduate Research Day Poster*, Apr. 2020.

Manuscript in Submission

- [M1]. R. Si, J. Song, **H. Sun**, and Z. An, “AutoNetSim: Intent-Driven Wireless Network Experimentation with Self-Evolving Agents”, submitted, ICNP 2026.
- [M2]. * L. Zhang, Z. Li, and **H. Sun**, “RF-PGS: Fully-structured Spatial Wireless Channel Representation with Planar Gaussian Splatting”, submitted, *IEEE Transactions on Communication*, 2025.
- [M3]. * K. Deng, H. Zhang, J. Lu, and **H. Sun**, “HoP: Homeomorphic Polar Learning for Hard Constrained Optimization”, submitted, *ICML*, 2025.

- [M4]. * P. Kudyba, et. al, “A UAV-assisted Wireless Localization Challenge on AERPAW”, submitted, *IEEE Communications Magazine*, 2025.

Patents

- [Pa1]. Anti-theft System Based on Power Line Carrier Communication and Dynamic Image Detection, China Patent Application: 201310421200.X. Publication Number: 103501188A, 2013.
- [Pa2]. Wiring-free indoor Wi-Fi access system based on power line carrier communications, China Patent Application: 201310069145.2, Publication Number: 103200700A, 2013.

Student Advising

• Graduate Students

- Lihao Zhang (PhD, UGA, Summer 2023); Proposal Defense Passed: Summer 2025, Expected graduation: Spring 2027.
- Hanwen Zhang (PhD, UGA, Fall 2023); Expected graduation: Spring 2028.
- Paul Kudyba (MS → PhD UGA, Spring 2026), College of Engineering Fellowship.
- Xiang Ma, distributed learning with wireless transmission schemes, and software defined radio applications on wireless sensing @USU (co-advising with Prof. Hu). Now Assistant Professor at UW-Eau Claire.
- Tilak Basnet, IoT Security and Privacy

• Undergraduate Students at UGA

- John Song, radio radiance field visualization
- Andrew Primiano, decentralized data infrastructure
- Austin Song, data valuation and incentives
- Nivedha Natarajan, database and frontend
- Sterling Strohauer, backscatter FM system
- Wayne Lam, iOS development

• Undergraduate Students at UWW

- Jacob Gasser, machine learning on edge devices and security
- Riley Redfern, machine learning on edge devices and security
- Shane Flynn, self-driving car and machine learning applications
- Kevin Marks, adaptive Bluetooth communication in IoT
- Dakota Vaughn, Mesh Bluetooth for landslide monitor.
- Christian Ayon, Mesh Bluetooth for landslide monitor.
- Jake Schnor, self-driving control algorithm implementation
- Colin Janaes, SaaS for IoT devices
- David Ascherl, SaaS for IoT devices
- Javin Herath, 60GHz Wi-Fi router data generation
- Brodie Peyer, digital steganography tool development

- JT Anderson, Medical IoT security and privacy
- Bryce Erdman, Medical IoT security and privacy
- Caitlin Seibel, landslide monitor software development
- Jeremy Angles, website and database design
- Brett Steck (Geology), landslide monitor data analysis
- Nick Huffman (Physics), landslide monitor software development
- Gabriel Lewis (Physics), landslide monitor software development

- **K-12 Students**

- Ethan Zhu (2019-2023), Green Caynon High School, Logan, UT. Now at UC Berkeley.

Student Achievements

- [S1]. Competitive Student Travel Grant: Paul Kudyba (INFOCOM 2024), Hanwen Zhang (ICC 2024), Lihao Zhang (ICC 2024).
- [S2]. Paul Kudyba (MS, UGA): NSF AERPAW “Find A Rover Challenge” 2nd and 3rd Place, INFOCOM 2024, Student Travel Grant.
- [S3]. Ethan Zhu (HS, Grand Cayon High School): ISICN 2024, Best Paper Award.
- [S4]. Jacob Gasser, Shane Flynn (BS, UWW): UWW Undergraduate Research Program Award, 2020.
- [S5]. Jake Schnor, UW Regent Scholar Undergraduate Researcher, 2021.
- [S6]. Tilak Basnet, UWW Graduate Research Award, 2021.

Conference/Invited/Seminar Presentations

- [T1]. “Radio Radiance Field: The New Frontier of Wireless Channel Representation”
Seminar talk to: NextG Channel Model Alliance, Feb. 2025; IEEE ISICN Tutorial Mar. 2025; Ericsson Research, May 2025; Samsung Research USA, Jul. 2025.
- [T2]. “HawkRover: An Autonomous Vehicular Communication Testbed with Multi-sensor Fusion and Deep Learning”, UW-Stout AI Club Seminar Talk, Dec. 2024.
- [T3]. “Driving the Future: Wireless Communication Solutions for Electrified Transportation”, *invited talk*, Asian American Engineer of the Year, New York, Sept. 2023.
- [T4]. “Approximate Wireless Communication for Federated Learning”, ACM WiSec Workshop, Guildford, United Kingdom, Jun. 2023.
- [T5]. “Efficient, Secure, and Intelligent Wireless Connectivity for Billions of Devices”, Idaho State University, Wayne State University, University of Houston, University of Georgia, University of Nebraska-Lincoln, University of North Carolina at Charlotte, Seminar Talk, Feb. 2022 - Apr. 2022.
- [T6]. “Open Radio Access Network and Learning Algorithms for Next-Generation Massive MIMO Applications”, Virtual presentation, IEEE Future Networks seminar talk, Jun. 2021.
- [T7]. “Towards Green Mobile Edge Computing offloading Systems with Security Enhancement”, Virtual presentation, IEEE i-ETC, Oct. 2020.
- [T8]. “High Precision Indoor Localization with Commodity Wi-Fi and Machine Learning”, Department colloquium seminar talk, Mar. 2020.

- [T9]. “A Field Study on Fingerprinting-based 28 GHz Localization”, Virtual presentation, IEEE SPAWC 2020.
- [T10]. “Non-orthogonal Multiple Access for 5G: Fundamentals, Challenges, and Design Considerations”, Invited talk, Qualcomm, Sony, University of Wisconsin-Whitewater, etc. Spring 2019.
- [T11]. “Robust Beamforming Design in Cognitive Radio Networks with SWIPT”, *IEEE ICC*, Kansas City, MO, May 2018.
- [T12]. “Challenges and Enabling Technologies in 5G Wireless Communications”, Invited talk, ECE colloquium, Utah State University, Nov. 2017.
- [T13]. “Outage Probability in a NOMA Relay System”, *IEEE WCNC*, San Francisco, CA, May 2017.
- [T14]. “Non-Orthogonal Multiple Access in Heterogeneous Networks”, *IEEE Globecom*, San Diego, CA, Dec. 2015.
- [T15]. “Introduction to Wireless Communication”, Guest speaker for undergraduate class “Academic Presentation”, Utah State University, Jan. 2015.

Teaching Experience

- **University of Georgia**
 ELEE 3270: Microelectronics I, Spring 2023, Spring 2024, Spring 2025.
 ELEE 8930: Optimization Theory and Engineering Application, Spring 2024, Spring 2025.
 ELEE 8990: Advanced Topic on Wireless Communication, Fall 2024, Fall 2025.
- **University of Wisconsin-Whitewater**
19 classes in 3 years, 6 different classes, developed 3 new courses, advised several capstone and thesis
 COMPSCI 174: Introduction to C++ Programming, 2 sections, Fall 2019, Spring 2020, Fall 2020, Fall 2021.
 COMPSCI 222: Intermediate C++, Fall 2019, Fall 2020.
 COMPSCI 456: Computer System Security, Spring 2020, Spring 2021. (**new course, evaluation: 4.13/5**)
 COMPSCI 412: Computer Organization and System Programming, Fall 2020, Spring 2021, Fall 2021.
 Cyber 742: Computer Forensics, Spring 2021. (**new graduate course, evaluation: 4.72/5**).
 Cyber 747: Embedded System Security (**new graduate course**, Spring 2022).
 COMPSCI 799: Thesis Research (Spring 2020, Fall 2020, Fall 2021).
 COMPSCI 498: Independent Study (Spring 2021, Fall 2021, Winter 2021).
- **Graduate Teaching Assistant, Utah State University**
 ECE/MAE 5310: Control Systems, Fall 2014.
 ECE 3410: Microelectronics I, Spring 2015.
 ECE 5600: Introduction to Computer Networks, Fall 2015, Fall 2016, Fall 2017.
 ECE 6600: Wireless and Mobile Networking, Spring 2016, Spring 2017.
- **Lab Session Assistant, Utah State University**
 ECE 3410: Microelectronics I, Spring 2015.
 ECE 5600: Introduction to Computer Networks, Fall 2015, Fall 2016, Fall 2017.

Professional Services

- *Editorial Board*
Associate Editor, IEEE Communications Magazine, Feb. 2024 - Present.
Editor, IEEE Networks Magazine, Nov. 2023 - Present.
- *Funding Agency Panel Review*
NSF CISE Core Program Panelist, 2024, 2025.
- *Workshop, Special Issue, Guest Editor*
Lead Organizer: 1st IEEE Workshop on UAV Communication and Experimentation (IEEE VTC 2024 Fall).
- *Technical Program Committee Chair*
IEEE WCSP 2023, Hangzhou China.
IEEE ISICN 2024, San Juan, Puerto Rico, USA.
- *Technical Program Committee Member*
IEEE ICNC 2025, ICC 2025, 2024, 2023, 2022, 2021, 2019: Advanced Mobile Edge/Fog Computing for 5G Mobile Networks and Beyond, Shanghai, China. IEEE INFOCOM 2022, 2023, 2024.
- *Reviewers (selected list, 200+ reviews)*
IEEE Transactions on Wireless Communications
IEEE Transactions on Communications
IEEE Transactions on Vehicular Technology
IEEE Journal on Selected Areas in Communications
IEEE Communications Letters
IEEE Signal Processing Letters
IEEE Internet of Things Journal
IEEE Wireless Communications Magazine
IEEE Communications Magazine
ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT)
IEEE Access
IEEE Vehicular Technology Magazine
Journal of Wireless Communications and Mobile Computing
KSII Transactions on Internet and Information Systems
Transactions on Microwave Theory and Techniques
International Journal of Electronics and Communications
Computer Communications
IEEE Global Communications Conference (Globecom)
IEEE International Conference on Communications (ICC)
IEEE Vehicular Technology Conference (VTC)
IEEE INFOCOM
Sensors
- *IEEE Future Networks Standardization*
Co-Chair in the Massive MIMO group (2018-2021)
Present in keynote and draft the roadmap documentations
Organize regular meetings, interact with other 8 groups
- *University and Department Services:*
UGA ECE Ad Hoc Graduate Program Committee (2024)
UGA ECE Curriculum Committee (2023, 2024)

UGA ECE Faculty Search Committee (2023)
UGA CS Faculty Search Committee (2023, 2024, 2025)
Cyber Security Program Coordinator (Engineering Emphasis), **landed B.S. and M.S. Cyber Security Program at UWW**
University Graduate Council Committee
College Inclusive Excellence Committee
College International Education Committee
Department Recruitment Committee
Department Colloquium Committee
Department Curriculum Assessment Committee
Department Faculty Search Committee (2021)
Department Student Thesis Committee (for several graduate students)
Faculty Academic Advisor: advise ~ 50 undergraduate students each semester

Media Coverage

- Team SunLab won NSF Advanced Wireless Platform AERPAAW “Find A Rover Challenge”, [Official NSF News](#), [News link](#).
- UW-Whitewater front page story, “Hands-on Research with an Impact, Making Self-driving Cars Smarter”. [News link](#).
- UW-Whitewater, “UW-Whitewater’s 2021 achievements and milestones”. [Video link](#). My lecturing and contribution to cyber security program appear in 2:12, my undergraduate research achievement appears in 3:23.
- Spectrum News 1, “UW-Whitewater offers cybersecurity bachelors degree”. [News link](#), contents include my sample course.
- The Gazette, “UW-Whitewater professor wins UW System’s regent scholar grant for second time”. [News link](#)
- The UW System, “2021 Regent Scholar Awards: UW System recognizes three professors for distinction in research, innovation, entrepreneurship”. [News link](#)
- UW-Whitewater News Room: “Assistant Professor in Computer Science Wins Prestigious Award”. [Video link](#)
- UW-Whitewater Cybersecurity Program Validated by the U.S. National Security Agency. [News link](#)